

PYTHON ESSENTIAL PROJECT

Minor Project Report

Branch / Subject: CSE

Academic Session: 2026–2027

1. PROJECT ABSTRACT

This project is a console-based arithmetic testing tool developed using Python. It provides an interactive terminal interface where users can choose mathematical operations like addition, subtraction, multiplication, and division. The application evaluates answers in real time, handles user selections, and calculates the total score. It is developed as a fundamental CSE programming lab project to demonstrate control structures and console I/O handling.

2. INTRODUCTION

Name: Pranav Parmar

Console-based tools are widely used to build foundational programming concepts. This project implements fundamental programming logic to test basic arithmetic abilities. The program prompts the user for their name, presents an operation-based menu, displays multiple-choice questions, validates answers, and gives instant performance feedback.

3. SYSTEM REQUIREMENTS

Programming Environment: Python 3.x (IDLE, VS Code, or Command Line)

Operating System: Windows 10/11, Linux, or macOS

Hardware: Any standard x86/64 system with minimum 2 GB RAM

4. IMPLEMENTATION & LOGIC FLOW

User Input: Takes player name via `input()` and initializes score counter to zero.

Menu Driven Selection: Reads numerical choice (1 to 4) using `int(input())` for arithmetic category selection.

Conditional Branching: Uses `if-elif-else` constructs to route user execution to the respective category.

Option Validation: Validates user input against pre-set multiple-choice answers using membership checking (`in`).

Result Display: Displays total score and session completion message upon quiz termination.

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- 1.

- 2.

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5. SOURCE CODE

Python Essential Project

```
name = input("Enter your name: ")
print("Welcome to the Python Essential Project,", name)
print("Test your calculation skills!")
print("\nChoose a topic:")
print("1. Addition")
print("2. Subtraction")
print("3. Multiplication")
print("4. Division")
user_input = int(input("Enter your choice (1-4): "))
score = 0
if user_input == 1:
    print(name, "here are the addition questions")
```

Question 1

```
print("1: 326 + 457 =")
print("(a) 773 (b) 783")
print("(c) 793 (d) 763")
n1 = input("Enter your answer: ")
if n1 == "b":
    print("Correct answer!")
    score = score + 1
elif n1 in ["a", "c", "d"]:
    print("Wrong answer!")
else:
    print("Invalid answer!")
```

Question 2

```
print("\n2: 648 + 275 =")
print("(a) 923 (b) 913")
print("(c) 933 (d) 943")
n2 = input("Enter your answer: ")
if n2 == "a":
    print("Correct answer!")
    score = score + 1
elif n2 in ["b", "c", "d"]:
```

```

print("Wrong answer!")
else:
print("Invalid answer!")
elif user_input == 2:
print("Subtraction quiz selected.")
elif user_input == 3:
print("Multiplication quiz selected.")
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elif user_input == 4:
print("Division quiz selected.")
else:
print("Invalid topic selection!")
print("\n=====")
print(name, "your score is", score)
print("=====")
print("Thank you for playing the Python Essential Project!")

```

6. EXECUTION OUTPUT

Enter your name:

Welcome to the Python Essential Project,

Test your calculation skills!

Choose a topic:

1. Addition
2. Subtraction
3. Multiplication
4. Division

Enter your choice (1-4): 1

here are the addition questions

1: $326 + 457 =$

(a) 773 (b) 783

(c) 793 (d) 763

Enter your answer: b

Correct answer!

2: $648 + 275 =$

(a) 923 (b) 913

(c) 933 (d) 943

Enter your answer: a

Correct answer!

=====

your score is 2

=====

Thank you for playing the Python Essential Project!

7. FUTURE SCOPE

Adding automated question generation using the random library.

Implementing try-except blocks for robust input handling.

Storing player scores using file handling or database connectivity.

Developing a graphical interface using Tkinter or Pygame.

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8. CONCLUSION

This minor project demonstrates the practical application of core Python programming concepts including menu-driven console flow, user input sanitization, dynamic scoring, and conditional evaluation.

9. REFERENCES

Python 3 Official Documentation (python.org)

Core Python Programming Reference Guides

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